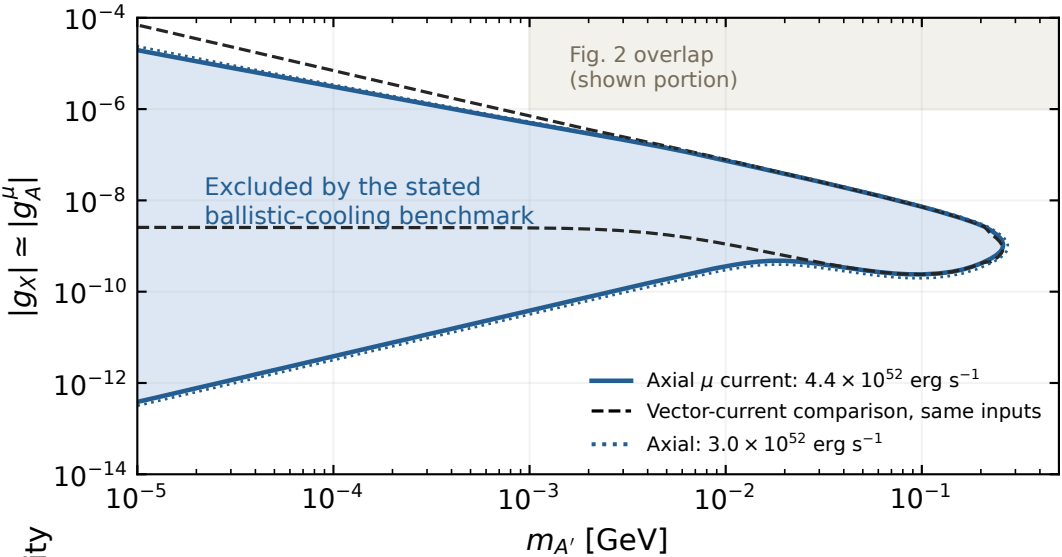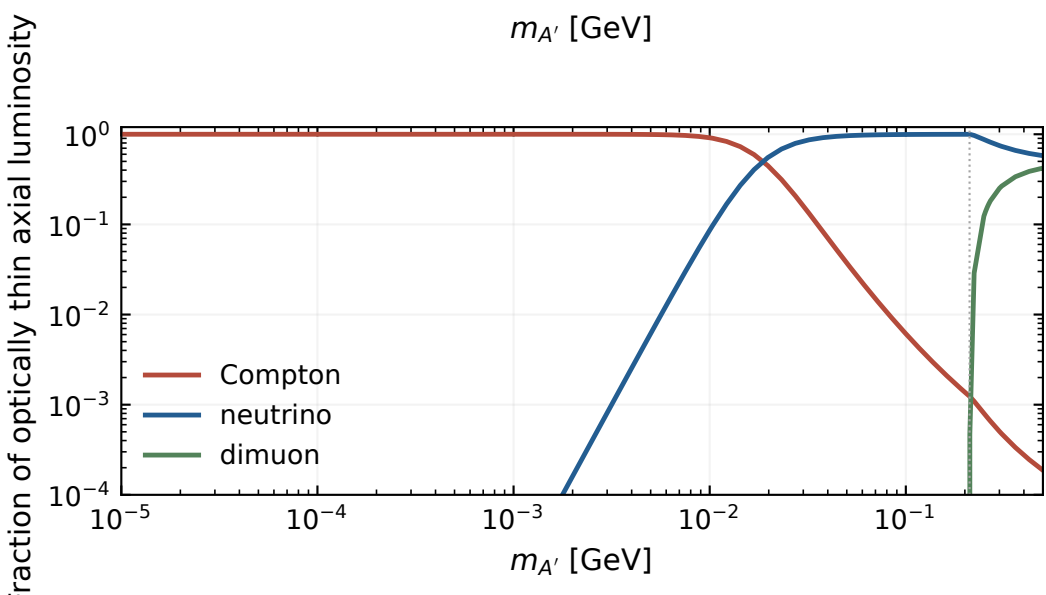SN1987A: axial-EFT fixed-profile benchmark (GeV)